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Sequence Bioinformatics **WS 2026/27**

Assignment 4 **Due: Wed 11-Nov-2026, 10:00**

The aim of this assignment is to investigate the gene trees of 78 chloroplast genes of water lilies and their relatives, with the methods of this week's lectures (maximum parsimony, models of sequence evolution and maximum likelihood, Chapter 4 of the handout, Sections 4.4 to 4.6) and of the coming one (support values, Section 4.8, Mon 09-Nov), and then to summarize the 78 gene trees by a consensus tree and a consensus network, which SplitsTree computes. It is a practical: you run the programs, report what they produce, and interpret it. For every step, say which tool and which parameters you used and what kind of method the tool implements, and hand in the commands you ran. Section numbers refer to the handout.

Setup

Download `assignment04.zip` from Ilias. It contains 78 files in FastA format, one per gene (`rbcL.fasta`, `matK.fasta`, ...), each with the DNA sequence of that gene from the same 17 chloroplast genomes, named by species and GenBank accession. You need an alignment program (MAFFT is recommended, `https://mafft.cbrc.jp`; Clustal Omega or MUSCLE will do), IQ-TREE (`http://www.iqtree.org`, version 2 or 3; the command is `iqtree2` or `iqtree3`), an alignment viewer (AliView, Jalview or SeaView, for instance) and SplitsTree (the SplitsTree App, `https://github.com/husonlab/splitstree6`). All run on Linux, macOS and Windows. Do not run 78 genes by hand: a shell loop of three lines does it, and the whole computation takes a few minutes.

1 Biological background (1 point)

The sequences come from this paper: M. Gruenstaeudl, *Why the monophyly of Nymphaeaceae currently remains indeterminate: an assessment based on gene-wise plastid phylogenomics*, Plant Systematics and Evolution 305, 827–836 (2019), `https://doi.org/10.1007/s00606-019-01610-5` (also on Ilias). Read the abstract and the introduction, and answer in a few sentences: What are the data, which plant families and genera do the 17 genomes represent, and which of them is the outgroup? What is the main question of the paper, and why does the genus *Nuphar* matter for it?

2 Multiple sequence alignments (2 points)

Compute a multiple sequence alignment for each of the 78 genes.

(a) Which tool did you use, with which parameters, and what type of approach does it implement (Chapter 2, Sections 2.4 and 2.5)?

(b) For every gene, report the number of sequences, the length of the alignment and the number of columns that contain a gap, as a table (a text file is fine). Which alignment is the shortest, which the longest, and how many genes align without any gap?

(c) These are protein-coding genes. Look at the lengths of the gaps in `rbcL` and `matK`: what do you expect for the length of an insertion or deletion in a coding sequence, and is that what you see? The alignment of `ycf1` has a gap in almost every column. Find out why, and name the sequences responsible.

3 Visualization of multiple sequence alignments (1 point)

Open the alignments of `rbcL` and of `ycf1` in an alignment viewer and explain what is shown: how the viewer displays residues, gaps and conservation, which regions of `rbcL` are conserved and which are variable, and what the picture of `ycf1` tells you about that gene. Include one screenshot of each.

4 Phylogenetic trees (2 points)

Compute a maximum-likelihood tree for each gene with IQ-TREE, letting it choose the model and computing 1000 ultrafast bootstrap replicates:

```
iqtree2 -s <gene>.aln.fasta -m MFP -B 1000 -prefix <gene>
```

(a) What type of approach is this (Section 4.6), how does IQ-TREE search for the tree (Section 4.6.2), and how does it choose the model (Section 4.6.3 and Exercise 4.13)?

(b) Collect the model that was selected for each gene (the line "Best-fit model according to BIC" of the `.iqtree` report) and tabulate how often each was chosen. For how many genes does the model include `+F`, `+G4` or `+I`, and what do these three components mean (Sections 4.5.4 and 4.5.5; `+I` is a proportion of invariable sites, which the handout does not cover: look it up)? How often is Jukes–Cantor the best model, and why does the choice fall on simpler models for some genes than for others?

(c) Draw the tree of `rbcL` with its bootstrap values (IQ-TREE writes it to `rbcL.treefile`; SplitsTree, Dendroscope or FigTree draw it). Which groups does it support strongly, which weakly? Is Nymphaeaceae, that is, *Nuphar* together with *Barclaya*, *Nymphaea*, *Victoria* and *Euryale*, a clade of this tree? Where does *Nuphar* go instead, and with what support?

5 Consensus tree (1 point)

A majority consensus tree of a set of trees on the same taxa contains exactly the splits (bipartitions of the taxa induced by the edges) that occur in more than half of the trees. Collect the 78 gene trees in one file and compute their majority consensus tree, with IQ-TREE (`iqtree2 -con -t all.treefile -minsup 0.5`) or in SplitsTree. Draw it, with the percentage of gene trees that contain each of its splits. Which groups do most gene trees agree on? Is Nymphaeaceae a clade of the consensus tree? Then count, for each of the three possible placements of *Nuphar*, how many of the 78 gene trees contain it: *Nuphar* with the other Nymphaeaceae, *Nuphar* with the Cabombaceae (*Brasenia* and *Cabomba*), or *Nuphar* with *Trithuria*. (Each placement is a split; the splits of the consensus network of Task 6, at a low threshold, come with their frequencies, or write a short script.) What do the counts say?

6 Consensus network (2 points)

A consensus network keeps the splits that occur in more than a given fraction (the threshold) of the trees, also those that contradict each other, and draws them as a network: where the trees agree it looks like a tree, and where they disagree it opens boxes whose sides are the conflicting splits. Open the file with the 78 trees in SplitsTree and compute a consensus network from them. Compute the network with the threshold at $\frac{1}{3}$ and then at a lower value such as 0.2. Which parts of the network are tree-like, and where does it open boxes? Which conflicting splits form the boxes around *Nuphar*, and how does the network change between the two thresholds? Does the picture agree with the title of the paper, and what would you answer to the question whether the Nymphaeaceae are monophyletic, on the basis of these 78 genes?

7 Sankoff's algorithm by hand (1 point)

In a maximum parsimony analysis of aligned DNA sequences, assume that a *transition* between `A` and `G`, or between `C` and `T`, costs 1, and a *transversion* between any two other different nucleotides costs 2. Given the values for the nodes v and w shown below, which values does Sankoff's algorithm (Section 4.4.3) compute for the node u with these costs? Show how each entry arises.

